

Git yourself a leg up *for student projects*

*Renee Noble
Cloud Advocate*

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Hi I'm Renee!

I like many things and
have many jobs!



Especially if they = Tech + Education + Community

I'm a Cloud Advocate at...



I get to tell cool people
how to do cool things
with tech!

...like I'll do now!

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But it doesn't stop there!

I also run a business

And a charity

And teach!



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I'm all about Tech + Education + Community

Today I've got
some **tech** for
your **teaching**
and your
community!



In this session

- Git: Friend of Foe
- Go shopping with Git
- GitHub tools for better projects

Many tools, one solution!

We'll cover a lot of tech today, but it all comes together!

GitHub

GitHub Pages

GitHub Templates

Codespaces

Copilot

GitHub Actions

Git: Friend of Foe?

Do you use Git/Github?

Did you find it tricky to
learn git at the start?

Do you want to teach Git
to students?

Do you think teaching Git
to students is hard?

My own Git journey

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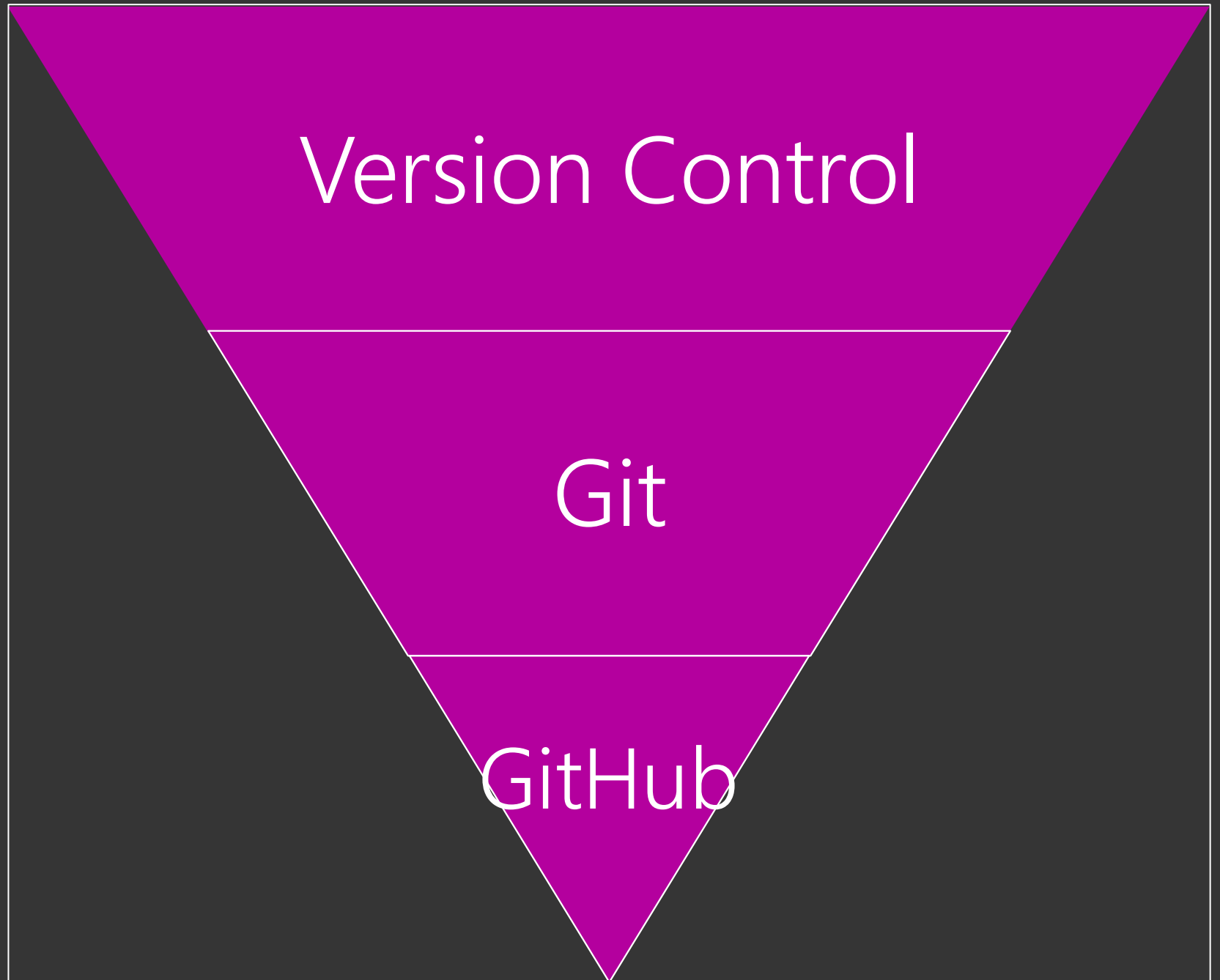




Let's take *Git* from the top!

Let's
talk
layers!

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Version Control

Systems that...

make sure your files don't look like **this**→

Includes ways to back up your files online, compare code, and more.

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Project draft



Project review



Project SUPER FINAL



Project final



Proect FINAL



Project DONE



Project DONE-5

Git

One implementation of version control

- Most common
- Uses branches
- Lots of integrations
- Lots of platforms that support it

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Git – How?

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/css/bootstrap.min.css"
7     integrity="sha384-TX8t27EcRE3e/ihU7zmQxVncDAy5uIKz4rEkgIXeMed4M0jlfIDPvg6uqKI2xXr2" crossorigin="anonymous">
8   <title>Translator</title>
9 </head>
10 <body>
11   <div class="container">
12     <h1>Translation service</h1>
13     <div>Enter the text you wish to translate, choose the language, and click Translate!</div>
14     <div>
15       <form method="POST">
16         <div class="form-group">
17           <textarea name="text" cols="20" rows="10" class="form-control"></textarea>
18         </div>
19         <div class="form-group">
20           <label for="language">Language:</label>
21           <select name="language" class="form-control">
22             <option value="en">English</option>
23             <option value="it">Italian</option>
24             <option value="ja">Japanese</option>
25             <option value="ru">Russian</option>
26             <option value="de">German</option>
27           </select>
28         </div>
29         <div>
30           <button type="submit" class="btn btn-success">Translate!</button>
31         </div>
32       </form>
33     </div>
34   </div>
35 </body>
36 </html>
```

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap@4.5.3/dist/css/bootstrap.min.css"
7     integrity="sha384-TX8t27EcRE3e/ihU7zmQxVncDAy5uIKz4rEkgIXeMed4M0jlfIDPvg6uqKI2xXr2" crossorigin="anonymous">
8   <link rel="stylesheet" href="../static/style.css">
9   <title>Translator</title>
10 </head>
11 <body>
12   <div class="container">
13     <h1>Write a letter to your penpal in your own language!</h1>
14     <div>Enter the text you wish to translate, choose the language, and click Translate!</div>
15     <div id="text-div">
16       <form method="POST">
17         <div id="submit-div">
18           <div class="form-group inline-form">
19             <label for="language">Language:</label>
20             <select name="language" class="form-control">
21               <option value="en">English</option>
22               <option value="it">Italian</option>
23               <option value="ja">Japanese</option>
24               <option value="ru">Russian</option>
25               <option value="de">German</option>
26             </select>
27           </div>
28           <div class="inline-form">
29             <button type="submit" class="btn btn-success">Translate!</button>
30           </div>
31         </div>
32         <div class="form-group" id="text-field">
33           <textarea id="textbox" name="text" cols="20" rows="10" class="form-control">
34         </div>
35       </form>
36     </div>
37   </div>
38 </body>
39 </html>
```

GitHub

A platform that supports Git projects

- Connects to your local files, to back them up online
- Make teams to share files/projects
- Tools work with others
- Tools to look at your code
- Integrates with other platforms



How can students understand all of this???

It's time for a shopping analogy!

Let's learn:

- Add
- Commit
- Push



Git == shopping

Code	Add	Commit	Push
------	-----	--------	------

Git == shopping

Code

Add

Commit

Push



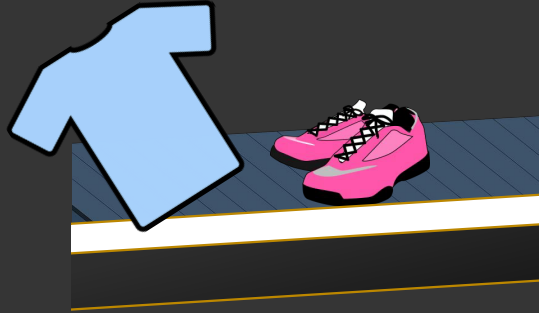
Git == shopping

Code

Add

Commit

Push



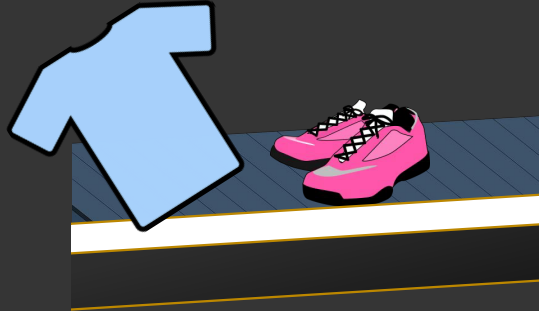
Git == shopping

Code

Add

Commit

Push

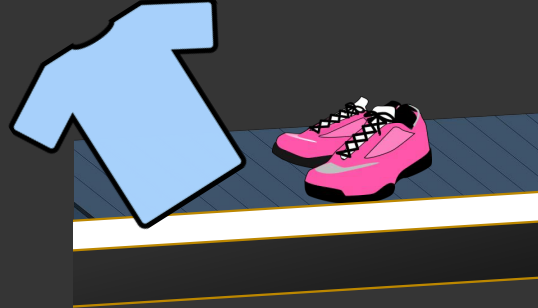


Git == shopping

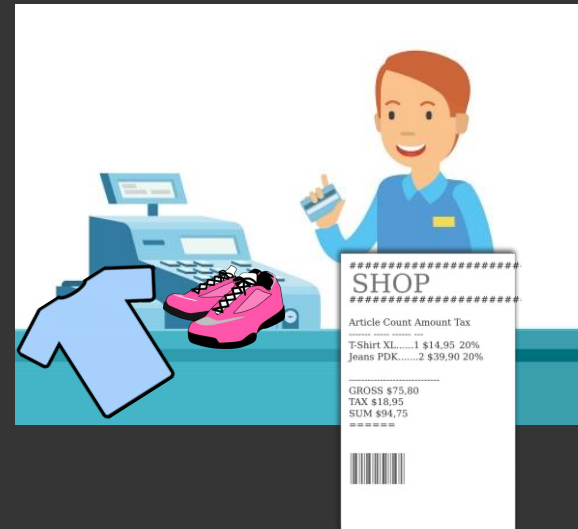
Code



Add



Commit



Push



Step 1 – Browse (code + save)

Your shopping trip starts with ...shopping!

Find things and put it in your trolley.

This is equivalent to you doing your coding in your projects. Git keeps track of the things you change.



Step 1 – Browse (code + save)



You can put things back on the shelf any time.

Do your **coding** like you normally do. Adding, removing, whatever you like. Putting something in your trolley is like **save**.



Step 2: At the register (add)

When you have some items you definitely want, you go to the register.

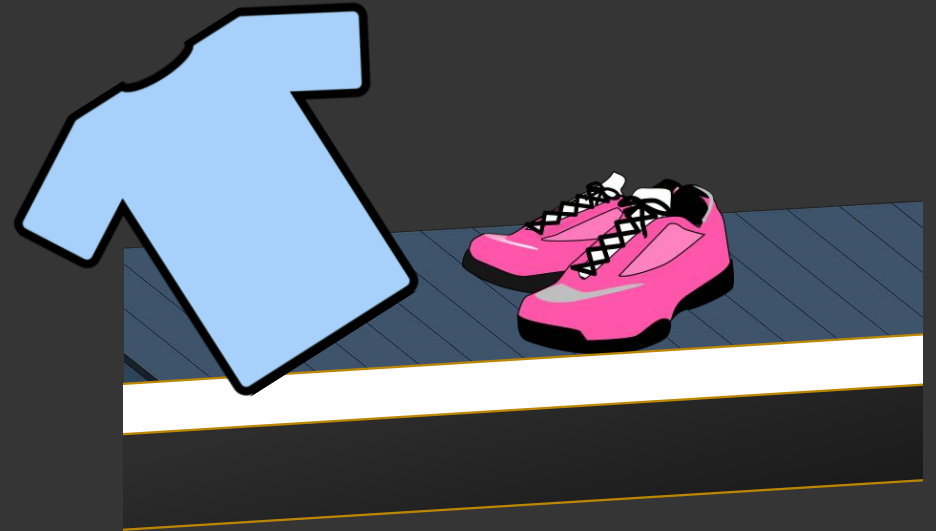
Put them on the conveyer belt!



Step 2: At the register (add)

You don't have to buy all of your items at once!

You can keep some in your trolley for later.
(Or to buy in a second transaction)

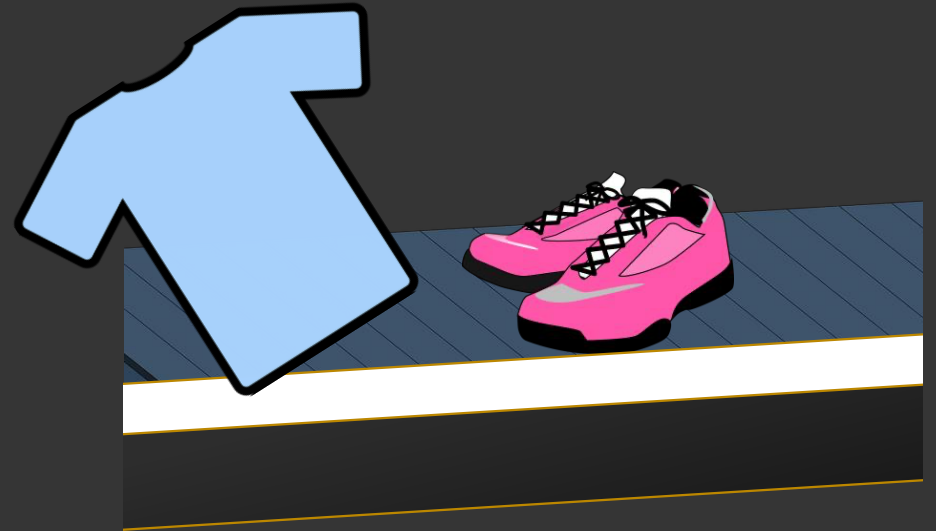


Step 2: At the register (add)

This step is like

git add

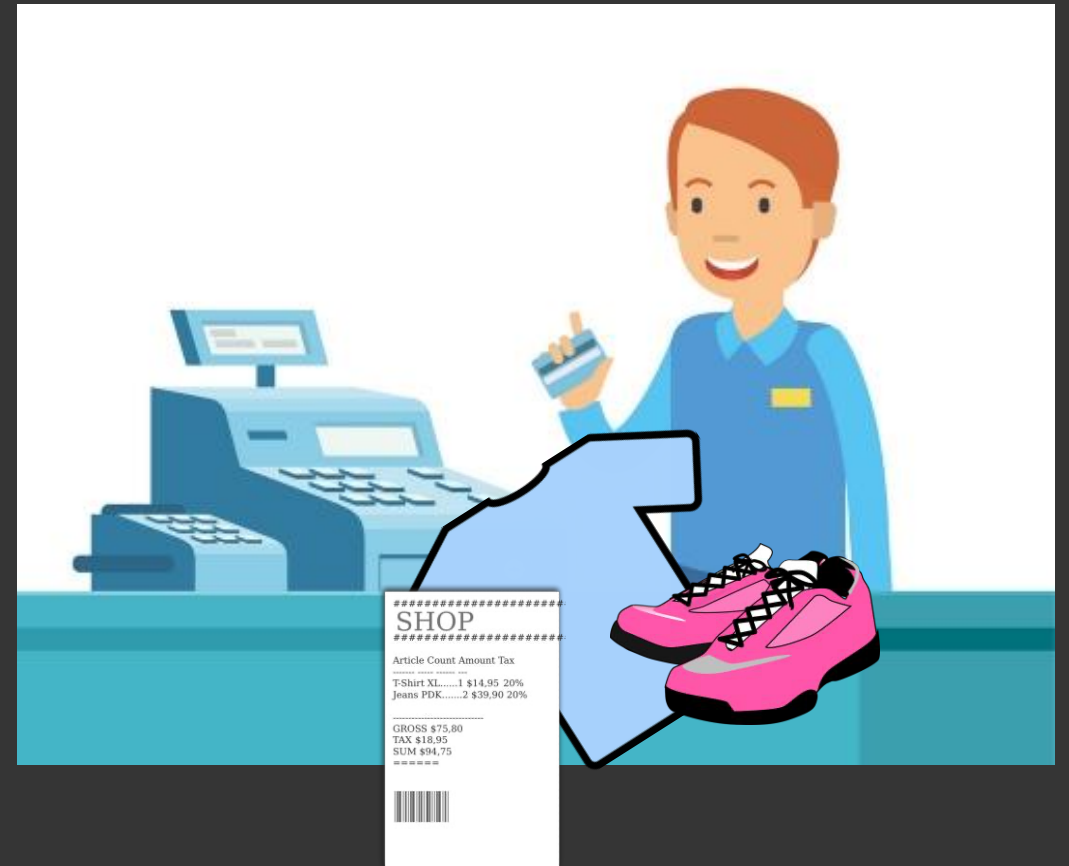
You can stage all the files (or parts of files) you want to commit in the next step.



Step 3 – Purchase (Commit)

When you've put through the items you want to buy, it's time to purchase!

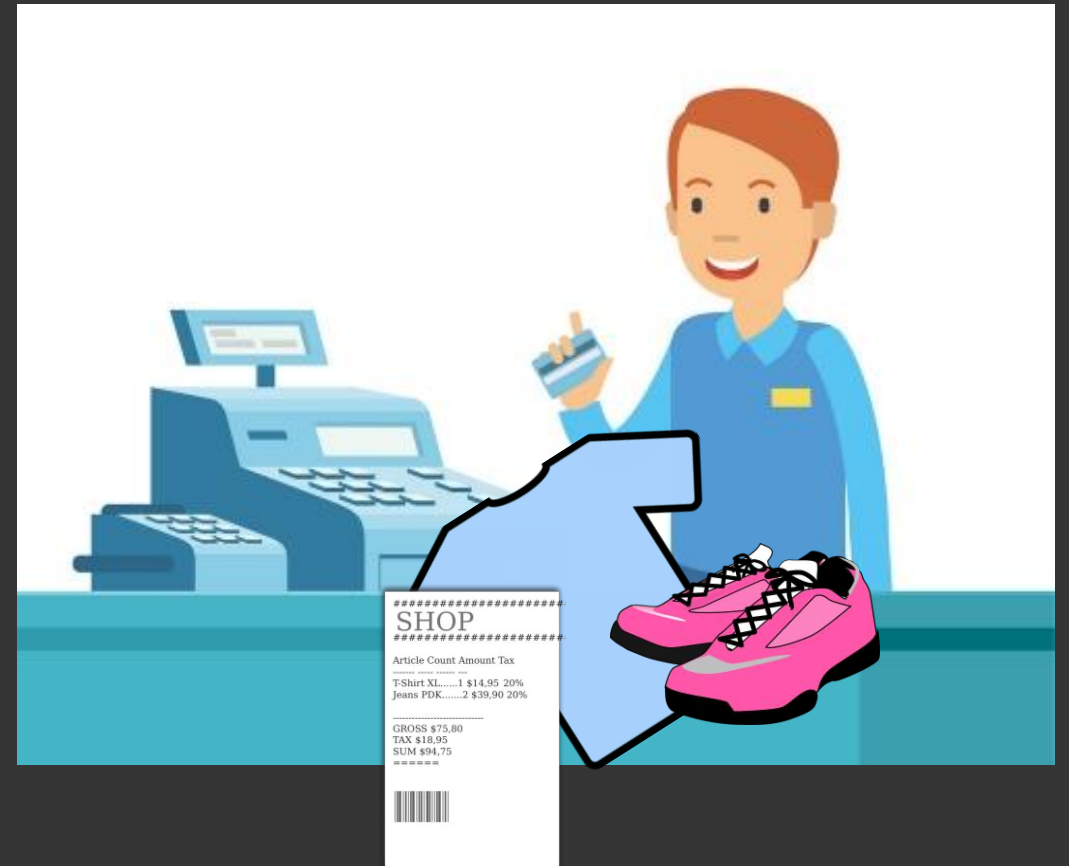
Before you purchase it's easy to say "actually no thanks to ... the shoes"



Step 3 – Purchase (Commit)

Completing the purchase creates a record/receipt.

If you need to undo the transaction you can.



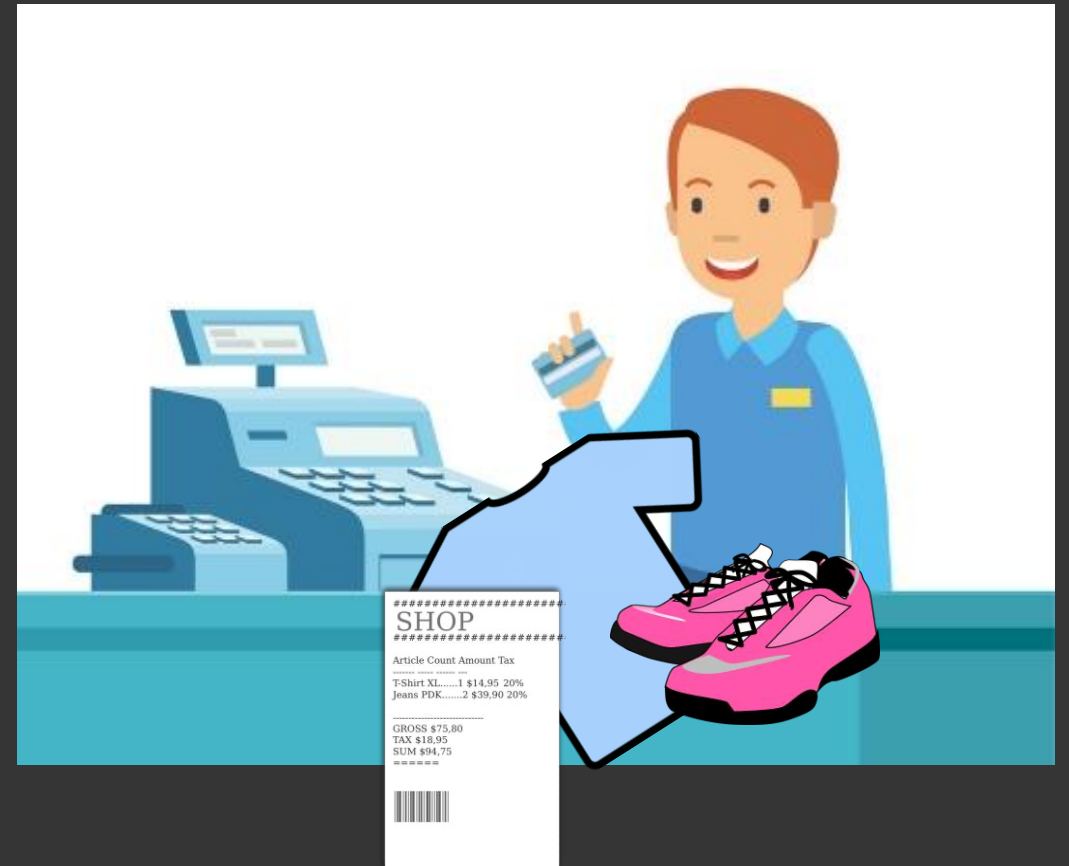
Step 3 – Purchase (Commit)

This step is like

git commit

You add the changes you made to the logs with:

```
git commit -m  
"<description of my  
commit>"
```



Step 4 – Instagram (Push)

When you want the rest of the world to know about your shopping trip... post it to social media!



Step 4 – Instagram (Push)

When you want the rest of the world to know about your shopping trip... post it to social media!

Wait until you've finished your shopping trip and post it all as one?



Step 4 – Instagram (Push)

This step is like

git push

If you're working alone, (no one follows your socials), not much can go wrong & your stuff is backed up

*If you're working in a team,
going back might be tricky.*



So now you know Git? What next?

Git is just the start for more projects!

I love storing stream code on GitHub, but also storing templates* to follow along with during streams.

[YOUR NAME]

[YOUR TITLE, or aspiring title]

[A BLURB ABOUT YOU AND YOUR GOALS] Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Contact me.

✉ email@example.com

👤 username

🐦 @twitter_handle

🌐 Your name on LinkedIn

My Projects

Add a screen shot of your project here.

[PROJECT TITLE]

[PROJECT DESCRIPTION] Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Technology

- Technology 1 (eg. Python)
- Technology 2 (eg. Flask)
- Technology 3 (eg. Azure Cognitive Services)

See it Live

GitHub Repo

Lots of engagement!

in-the-code-garden / portfolio-template

<> Code

Issues

Pull requests 2

Actions

Projects

Wiki

Security

portfolio-template Public

Edit Pins

Unwatch 1

Fork 31

Star 3

main

Go to file

Add file

<> Code

About

Branches

Tags

Get started with your GitHub Pages portfolio by cloning this

index.html

index.html > html

1<html>

2

3<head>

4<link href="style.css" r

5<title>Renee Noble - Por

6<link rel="stylesheet" h

7<link rel="preconnect" h

8<link rel="preconnect" h

9<link href="https://font

10</head>

11

12<body>

13

14

style.css

style.css > body

1body {

2font-family: 'Imprima', sans

3

4font-size: 12px;

5max-width: 960px;

6margin: auto;

7}

8

9#header-main {

10display: grid;

11grid-template-columns: 70% 3

12margin-top: 3em;

13}

14

CodeSwing

Dear Friend,

COME ALONG AND HAVE A GREAT DAY at...

Birthday party!

Date: 37th Mocktober

Time: 3pm

See you there!

Renee

Technology

• Python

• Flask

• Azure App Service

• Azure Blob

See it Live

GitHub Repo

Language Translation Web App

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Technology

• Python

• Flask

• Matplotlib

• Azure Machine Learning Studio

• NLP

GitHub Repo

Ln 1, Col 1

Spaces: 4

UTF-8

LF

HTML

Layout: U.S.

Video of final github page

My stream viewers loved it!

What about students?

My stream viewers loved it!

What about students?

It was met with some
confusion...



My stream viewers loved it!

What about students?

It was met with some confusion...

But with some new tools, there's a new way!



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reene noble.com

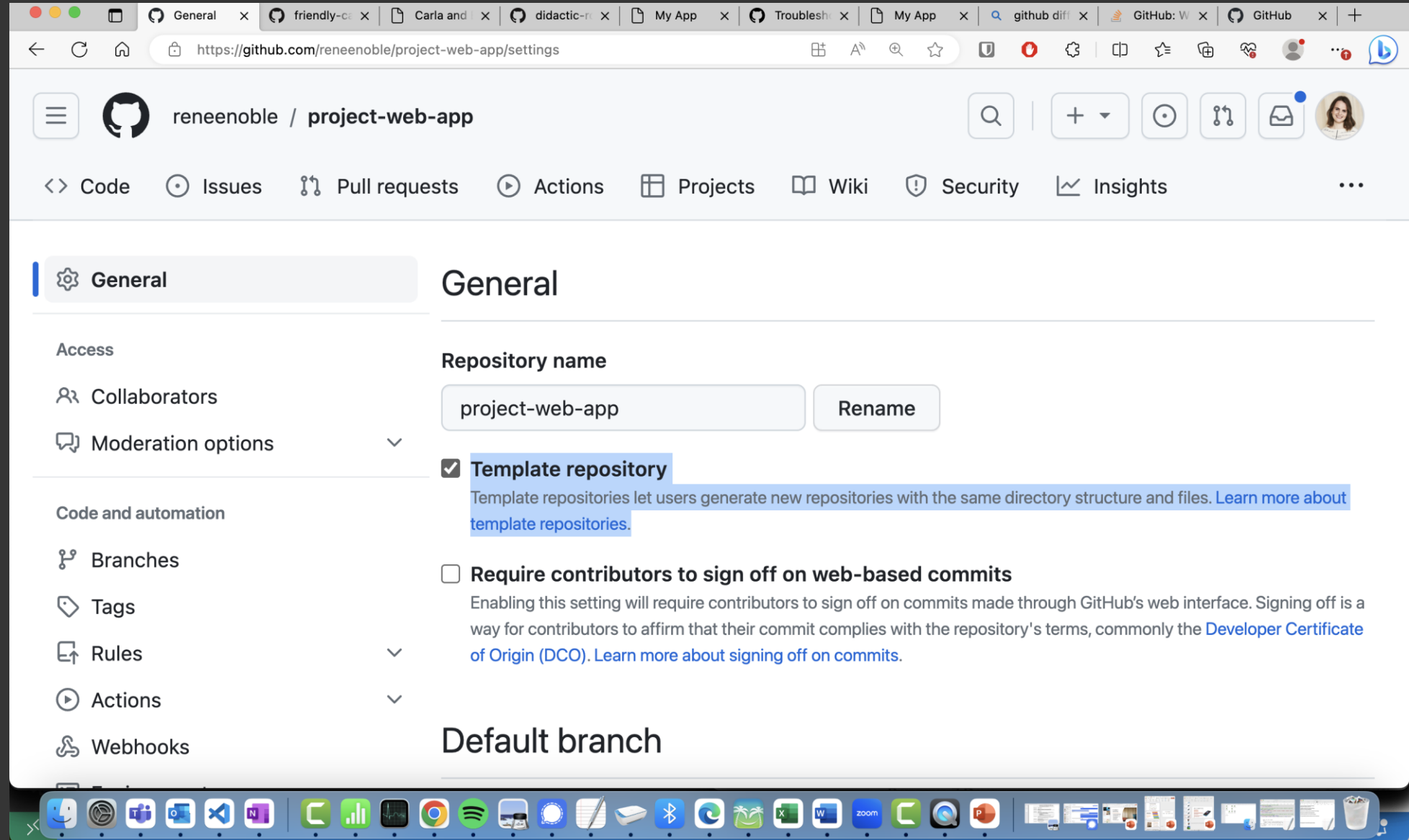
An easy little win!

GitHub templates

Lose the commit history!

Loose the upstream concerns

"Fork" you're own



Let's make a better
project scaffold!

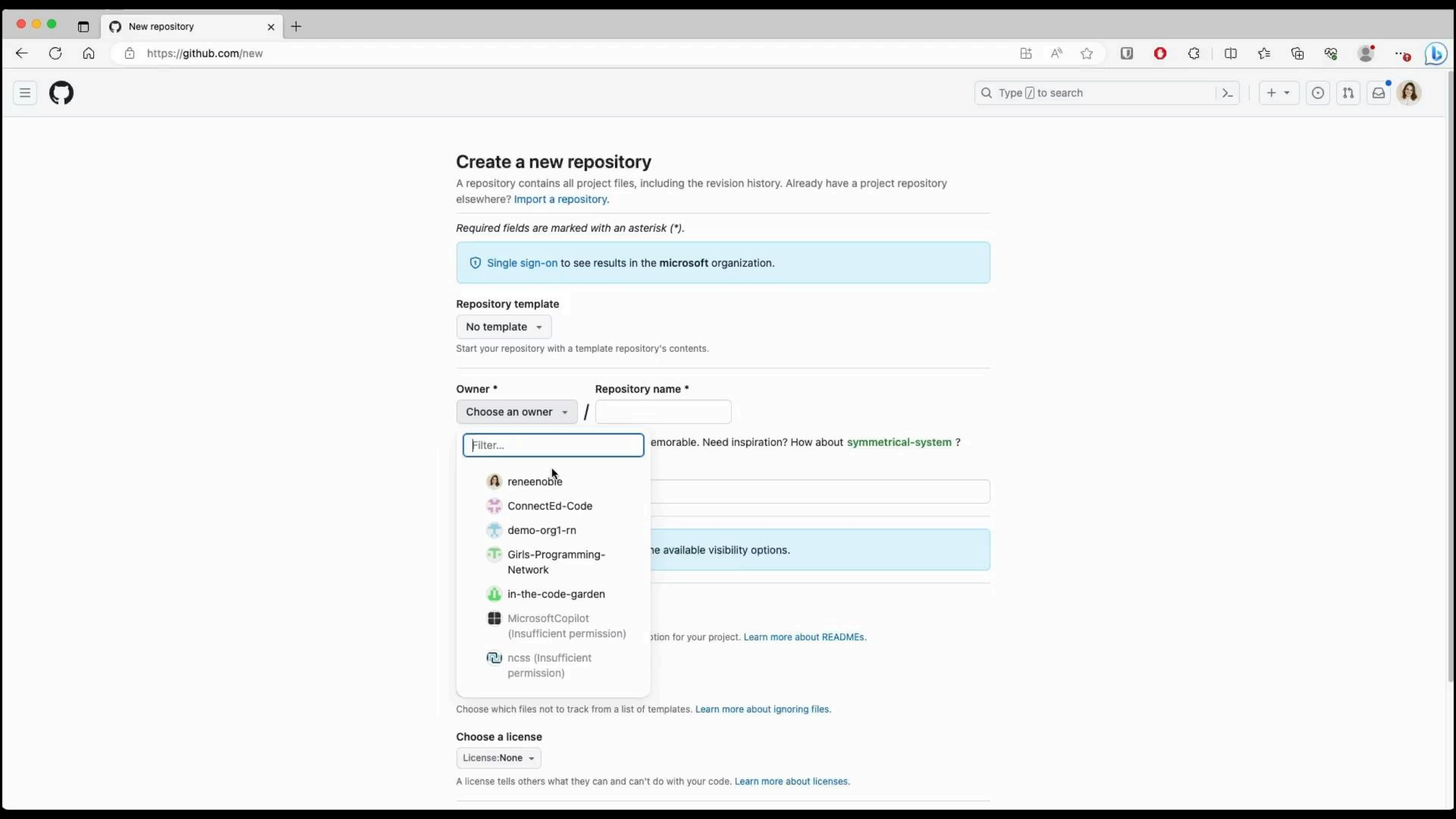
We'll use it all!

- GitHub
- GitHub Templates
- Codespaces
- Copilot (for us!)
- GitHub Actions



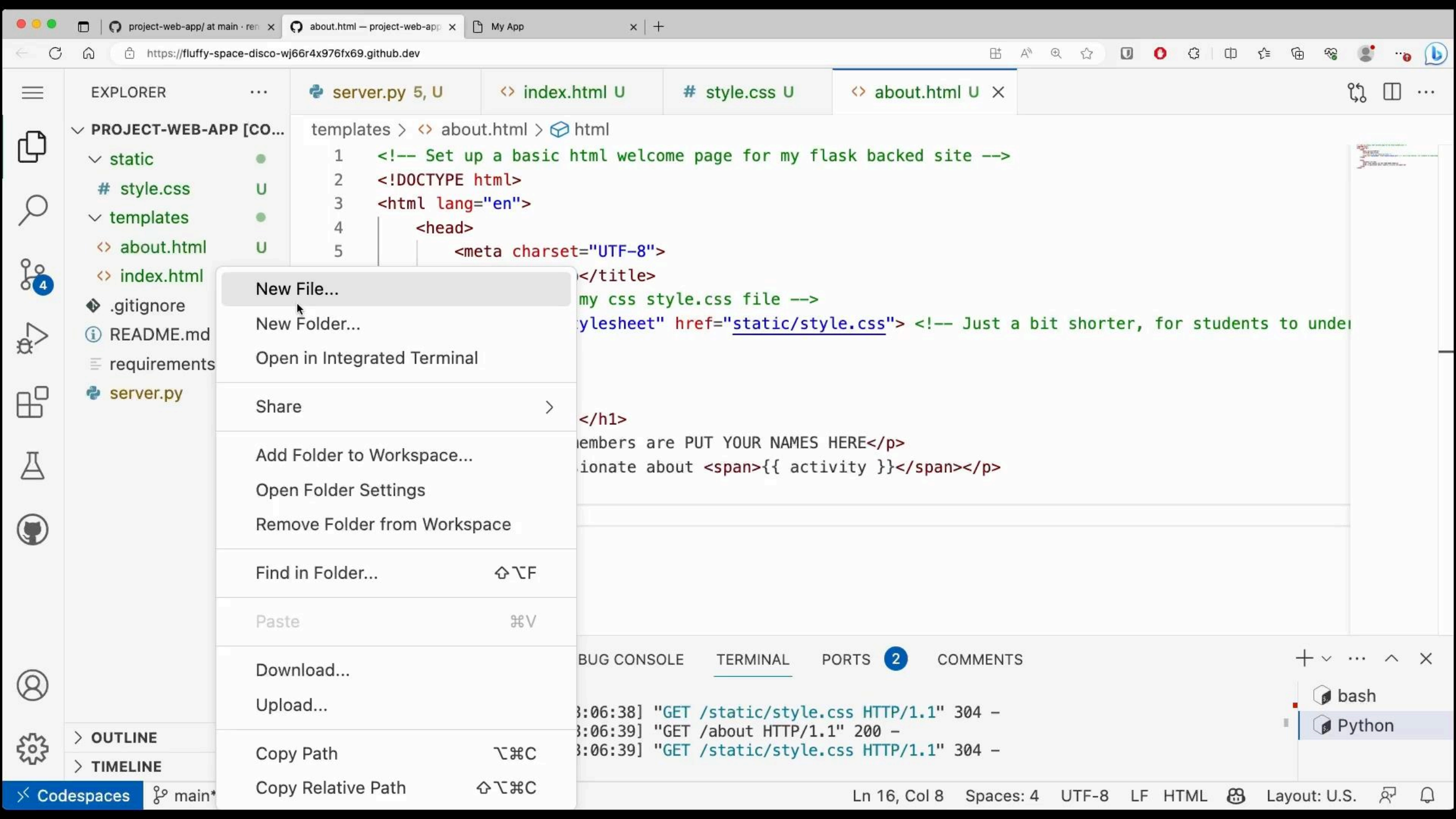
Speed run making a flask app!

Thanks to Codespaces and Copilot!



Adding testing with ease!

Thanks to Copilot, VS Code Flask tools, and GitHub Actions



How your *students* will see it in action

Thanks to GitHub templates

Take Aways

Git, GitHub and many tools...

are friend to you and
your students!



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Take Aways

Git doesn't have to seem so scary...

if you treat it like a shopping trip!

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Take Aways

Just a little git set up...

Can *git* you and your students a leg up in the classroom!

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More stuff for you and your students!

Check out my website for...

Aka.ms/renee-pycon2023

- A Cloud Skills Challenge
- Videos on Git, Python, and web dev
- Links to Imagine Cup and Imagine Cup Junior
- Links to Microsoft Learn Student Ambassadors
- Links to Founders Hub



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